

Jermaine Tyrell Dennis

U.S. Citizen · Based in California (PST) · Fully Remote

Phone: 323-342-7069 · Email: jermainedennis@live.com

Portfolio: jermainedennis.wixsite.com/software-engineering

Summary

AI Engineer and Full-Stack Architect with 10+ years shipping production software across gaming (Square Enix – Final Fantasy, Kingdom Hearts), film VFX (Method Studios – Marvel), and enterprise retail (Home Depot, Lowe's). Solo-architected and shipped a cross-surface AI agent platform with real-time voice (3 pipelines), autonomous agent networking, RAG memory, MCP protocol integration, GraphQL subscriptions, and full billing – serving real users across web, phone, Instagram, LINE, website embeds, and XR. Deep experience with OpenAI APIs, real-time WebSocket systems, vector search (pgvector), and production AI application design.

Core Skills

AI & LLM Integration: OpenAI Realtime API, Gemini Live, Groq, OpenAI Embeddings, RAG pipelines, vector search (pgvector + HNSW), trust-aware retrieval, prompt engineering, autonomous agent orchestration

Backend: Python, Django (async), Strawberry GraphQL, PostgreSQL, Redis, WebSocket (channels), Celery, REST & GraphQL APIs

Frontend: TypeScript, React 19, Next.js 15 (App Router), Tailwind CSS, Three.js, WebGL, Babylon.js, Unity/C#

Protocols & Integrations: MCP (Model Context Protocol – bidirectional), Google A2A Agent Cards, Twilio (voice + SMS), Stripe, PayPal, Instagram Graph API, LINE Messaging API, OAuth 2.0, HMAC-signed webhooks

Infrastructure: Docker, CI/CD, ASGI, WebSocket auth (multi-token), cron orchestration, Render, AWS

Real-Time Systems: GraphQL subscriptions (graphql-ws), WebRTC (OpenAI Realtime direct mode), pub/sub event pipelines, server-sent animation events

Experience

Hololoop Inc. — Founder / AI Engineer

Remote · 2022 – Present

Architected and solo-built **Yakimo**, a production AI agent platform where users create animated AI personas that handle networking, conversations, and scheduling across 7+ channels.

AI & Voice Systems:

- Built 3 production voice pipelines: OpenAI Realtime API (WebRTC direct + server sideband), Cartesia Cascaded (Twilio ↔ WebSocket), and Gemini Live — each with RAG injection, memory persistence, and credit enforcement
- Designed trust-aware RAG retrieval system: pgvector embeddings, privacy-level inheritance, knowledge categorization, and ACL-scoped search with lexical fallback
- Built autonomous agent discovery engine: vector similarity matching on agent profile embeddings, double opt-in connection protocol, staggered cron scheduling, and LLM-scored compatibility

Platform & Integrations:

- Implemented MCP bidirectional protocol — every agent is both an MCP server (callable by OpenClaw/n8n/Zapier) and MCP client (calls external tools during conversations). Published to ClawHub skill registry
- Shipped Google A2A Agent Card endpoint for protocol-level agent discovery
- Built social feed system: polymorphic posts/comments/likes, semantic + keyword search (pgvector cosine on post embeddings), connection requests, and agent profile directory with vector search
- Integrated Instagram DM, LINE Messaging, Twilio phone/SMS, and embeddable website chat widget (floating + inline modes with postMessage bridge)

Billing & Monetization:

- Built full billing stack: Stripe + PayPal checkout, 3 subscription tiers, three-layer creator pricing (platform → markup → agent), credit enforcement across all channels, creator payout (PayPal + bank transfer), revenue analytics dashboard

Frontend & Real-Time:

- Built Next.js 15 frontend with React 19: full-screen animated agent chat, GraphQL subscriptions for real-time typing/presence/party mode, WebSocket auth with dual-token fallback (bearer + guest), infinite scroll message pagination

- Designed character interaction system: device actions (poke/shake/spin/rub) with AI response pipeline + instant audio, external-aware actions (phone calls and DMs trigger real-time character animations via pub/sub)

Validated Distribution:

- Shipped GDC 2026 Event Finder as utility-first micro-tool — 10+ organic user conversations via LinkedIn/Reddit distribution. Frontend MCP Bridge coordinated agent ↔ UI for interactive event filtering
 - 97/100 QR scan rate at SF Techweek (pre-animation system) — validated IRL acquisition mechanic
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3DCloud / Programming.com — Frontend 3D Software Engineer (Contract)

Remote · 2025

- Built web-based 3D product configurators used by Home Depot, Lowe's, and Howdens
 - Enhanced 3D modeling workflows and debugged enterprise-scale systems for stability and performance
 - Technologies: React, TypeScript, C#, WebGL, real-time APIs
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Contract Software Development and Consultancy

Los Angeles, CA · 2018 – 2019

- Various software development projects
 - Developed Python automation scripts for lighting and asset workflows
 - Helping companies make impactful technology decisions
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Square Enix (Japan) — Pipeline Tools Engineer / Lighting TD

Tokyo, Japan · 2016 – 2018

- Built Python automation tools and asset management pipelines for Final Fantasy and Kingdom Hearts
 - Developed front-end interfaces for 3D asset and render management
 - Collaborated with international teams; lived and worked in Japan
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Self-Published AR/VR Titles — Meta Start Developer

Remote · 2021 – 2023

- Developed and published VR applications for Oculus and Apple platforms (Unity/C#)
 - Built interactive 3D environments with AI-driven characters and real-time physics
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Selected Projects

Yakimo – AI Agent Platform (Production, live users)

Cross-surface AI agent platform. 3 voice pipelines, autonomous discovery, MCP bidirectional, social feed with vector search, A2A Agent Cards, 7+ channel delivery, full billing. Solo-built.

GDC 2026 Event Finder (Production, deployed)

Embeddable AI-powered event discovery tool. CSV → RAG ingestion → Yakimo agent → embed widget with Frontend MCP Bridge for interactive UI coordination. 10+ real user conversations.

Telepresence VR Prototype

Unity/C# client + Django backend; real-time interaction via WebSockets; optimized for low-latency immersive communication.

Education

- **BFA, Illustration** – University of the Arts, Philadelphia, PA
- **Certificate, Computer Graphics** – Gnomon School of Visual Effects, Hollywood, CA